

- 388.** There are some boys and girls in a room. The square of the number of girls is less than the square of the number of boys by 28. If there were two more girls, the number of boys would have been the same as that of the girls. The total number of boys and girls in the room is (C.P.O., 2007)
- (a) 7 (b) 10
(c) 14 (d) 56
- 389.** A classroom has equal number of boys and girls. Eight girls left to play kho-kho, leaving twice as many boys as girls in the classroom. What was the total number of girls and boys present initially?
- (a) 16 (b) 24
(c) 32 (d) Cannot be determined
(e) None of these
- 390.** What is the value of the expression $(1+x)(1+x^2)(1+x^4)(1+x^8)(1-x)$?
- (a) $x^8 + 1$ (b) $x^{16} - 1$
(c) $1 + x^{16}$ (d) $1 - x^{16}$
- 391.** The expression $\frac{1}{x-1} - \frac{1}{x+1} - \frac{2}{x^2+1} - \frac{4}{x^4+1}$ is equal to
- (a) $\frac{8}{x^8+1}$ (b) $\frac{8}{x^8-1}$
(c) $\frac{8}{x^7-1}$ (d) $\frac{8}{x^7+1}$
- 392.** $\left(x + \frac{1}{x}\right)\left(x - \frac{1}{x}\right)\left(x^2 + \frac{1}{x^2} - 1\right)\left(x^2 + \frac{1}{x^2} + 1\right)$ is equal to (S.S.C., 2006)
- (a) $x^6 - \frac{1}{x^6}$ (b) $x^8 - \frac{1}{x^8}$
(c) $x^6 + \frac{1}{x^6}$ (d) $x^8 + \frac{1}{x^8}$
- 393.** If $\left(x + \frac{1}{x}\right) = 3$, then $\left(x^2 + \frac{1}{x^2}\right) = ?$
- (a) 7 (b) 9
(c) 10 (d) 27
- 394.** If $\left(a - \frac{1}{a}\right) = p$, then $\left(a^2 + \frac{1}{a^2}\right)$ is equal to
- (a) p (b) $p - 2$
(c) $p + 2$ (d) None of these
- 395.** If $\frac{x-1}{x} = 3$, then the value of $1 + \frac{1}{x^2}$ is
- (a) 9 (b) 10
(c) 11 (d) None of these
- 396.** If $\left(x^2 + \frac{1}{x^2}\right) = 34$, then $\left(x + \frac{1}{x}\right)$ is equal to
- (a) 3 (b) 4
(c) 5 (d) None of these
- 397.** If $\left(x + \frac{1}{x}\right) = 2$, then $\left(x - \frac{1}{x}\right)$ is equal to (R.R.B., 2006)
- (a) 0 (b) 1
(c) 2 (d) 5
- 398.** If $\left(a + \frac{1}{a}\right) = 6$, then $\left(a^4 + \frac{1}{a^4}\right) = ?$ (R.R.B., 2008)
- (a) 1154 (b) 1158
(c) 1160 (d) 1164
- 399.** If $\left(x - \frac{1}{x}\right) = 2$, then the value of $\left(x^4 + \frac{1}{x^4}\right)$ is
- (a) 4 (b) 8
(c) 12 (d) 34
- 400.** If $\left(x - \frac{1}{x}\right) = \sqrt{21}$, the value of $\left(x^2 + \frac{1}{x^2}\right)\left(x + \frac{1}{x}\right)$ is
- (a) 42 (b) 63
(c) 115 (d) 120
(e) 125
- 401.** If $\left(x^4 + \frac{1}{x^4}\right) = 322$, the value of $\left(x - \frac{1}{x}\right)$ is
- (a) 4 (b) $3\sqrt{2}$
(c) 6 (d) 8
- 402.** If $x + \frac{1}{x} = p$, the value of $x^3 + \frac{1}{x^3}$ is
- (a) $p^3 + \frac{1}{p^3}$ (b) $p^3 - 3p$
(c) p^3 (d) $p^3 + 3p$
- 403.** If $x - \frac{1}{x} = 1$, then the value of $x^3 - \frac{1}{x^3}$ is equal to
- (a) 2 (b) 3
(c) 4 (d) 8
- 404.** If $\left(a^4 + \frac{1}{a^4}\right) = 1154$, then the value of $\left(a^3 + \frac{1}{a^3}\right) = ?$
- (a) 198 (b) 200
(c) 216 (d) None of these
- 405.** If $\left(x + \frac{1}{x}\right) = 3$, then the value of $\left(x^6 + \frac{1}{x^6}\right)$ is
- (a) 322 (b) 364
(c) 414 (d) 927
- 406.** If $\left(x + \frac{1}{x}\right) = \sqrt{13}$, then the value of $\left(x^3 - \frac{1}{x^3}\right)$ is (Delhi Police, 2010)
- (a) 26 (b) 27
(c) 30 (d) 36
- 407.** If $\left(4b^2 + \frac{1}{b^2}\right) = 2$, then $\left(8b^3 + \frac{1}{b^3}\right) = ?$ (S.S.C., 2008)
- (a) 0 (b) 1
(c) 2 (d) 5